

Veritas Report

1. Configurations

We executed three configurations in the Veritas framework:

1. (DESIGN, TYP) = (adder, 3-bit)
 2. (DESIGN, TYP) = (decoder, 2x4)
 3. (DESIGN, TYP) = (adder, 5-bit)
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2. Run Details

Run 1: adder (3-bit)

Inputs:

A0, A1, A2, B0, B1, B2, Cin

Outputs:

S0, S1, S2, Cout

Generated files:

adder_3-bit.cnf
adder_3-bit.bench
adder_3-bit.v
adder_3-bit_tab.csv

Run 2: decoder (2x4)

Inputs:

A0, A1

Outputs:

Y0, Y1, Y2, Y3

Generated files:

decoder_2x4.cnf
decoder_2x4.bench
decoder_2x4.v
decoder_2x4_tab.csv

Run 3: adder (5-bit)**Inputs:**

A0, A1, A2, A3, A4, B0, B1, B2, B3, B4, Cin

Outputs:

S0, S1, S2, S3, S4, Cout

Generated files:

adder_5-bit.cnf
adder_5-bit.bench
adder_5-bit.v
adder_5-bit_tab.csv

3. Oracle Validation Results

For each configuration, we implemented a Python oracle to compute the expected outputs and compared them against the generated truth tables (*_tab.csv).

adder (3-bit)**Result: PASS**

All rows in adder_3-bit_tab.csv matched the oracle output.

decoder (2x4)**Result: PASS**

All rows in decoder_2x4_tab.csv matched the oracle output.

adder (5-bit)

Result: **PASS**

All rows in `adder_5-bit_tab.csv` matched the oracle output.

4. Reflection

Conjunctive Normal Form (CNF) is a useful intermediate representation for correctness-by-construction because it provides a formal and compositional way to describe Boolean logic. Each clause encodes a constraint that must be satisfied, and the conjunction of clauses ensures that all logical relationships are enforced simultaneously. This makes CNF particularly suitable for verification, since it can be directly evaluated, checked for satisfiability, and systematically composed across modules. In the Veritas framework, representing circuits in CNF enables the automatic generation of correct-by-construction logic that can be validated through truth tables and further transformed into BENCH and Verilog formats. The structured nature of CNF also makes it easier to debug errors, as incorrect behavior can often be traced back to specific clauses.